

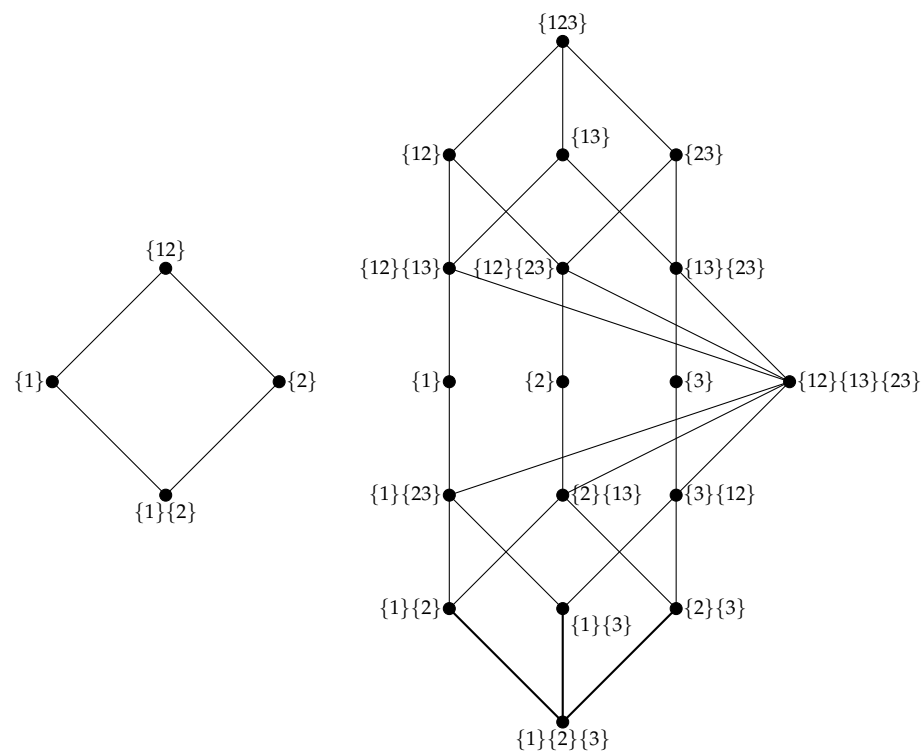


Figure 2. Partial information lattices. On the left is the lattice for two predictor variables, and on the right is the lattice for three predictor variables. Each lattice is constructed and annotated following the notation in [1]. Lattice vertices are partial information atoms, i.e., the unique modes of information-sharing that comprise the overall joint mutual information. Information atoms are denoted by index only: for example, $\{1\}\{2\}$ is the information redundantly disclosed by X^1 or X^2 , $\{1\}\{23\}$ is the information disclosed by X^1 or (X^2 and X^3), etc. Lattice edges indicate which atoms subsume other atoms. Atoms connected to and below other atoms consist of components of and/or subsets of the higher atoms, for example, $\{1\}\{2\} \preceq \{1\}\{23\}$ and $\{2\}\{13\}$, since information disclosed by $\{1\}\{2\}$ would also be visible to $\{1\}\{23\}$ and $\{2\}\{13\}$ if we did not use the Möbius inversion.

4.3. Choosing a Redundancy Measure

Applying PID analyses to real data requires some notion of redundant information ($I_{\cap}(\cdot)$) to be operationalized. To date, no single universally accepted measure has been proposed. All measures have their own trade-offs and drawbacks (such as only being defined for systems of a fixed size, only being amenable to discrete random variables, or requiring arbitrary thresholds). There are, at present, close to a dozen competing redundancy functions (see [22,29,31–40]). In the absence of a single accepted measure, different contexts may require the choice of different functions. For example, having more than two predictors precludes the measures proposed in [24,31], while continuous data cannot be analyzed with the measures proposed in [24,40], and so on. The majority of the studies discussed in this paper used the original $I_{\min}$ measure proposed in [1], although this measure has been criticized for unintuitive behavior [29,39]. For a deeper discussion of the practical considerations, see Section 6.

5. PID in Action

We next illustrate the application of information decomposition to the *empirical study* of neural information processing in biological circuits. This is not intended to be an exhaustive review of relevant studies. Rather, the goal is to offer a crisp window into the types of insights that can emerge. We draw preferentially on our own work out of convenience, but this work in turn was inspired by, and complements, other exemplary lines of empirical work (for examples, see [5,21,35,41–46]).